



Dalhousie University

Department of Psychology

Eighteenth Annual
In-House Convention

9-10 September 1992

SCHEDULE

Wednesday, September 9

9:00— 9:15 Coffee

9:15— 9:30 Opening Remarks: J. Clark

9:30—11:00 Session I. Chair: C. Beaver

- 1) P. McGrath, G.A. Finley, G. McNeill, P. Forward and P. Fitzgerald
- 2) K.W. Jensen and J.C. Fentress
- 3) J. Hamm

COFFEE BREAK

11:30— 1:00 Session II. Posters (Lounge)

- 4) J. Ryon, P. Paquet and R. Brown
- 5) H. Schellinck
- 6) A. D'Aloisio, M. Sullivan and F. Murphy
- 7) J. Francis and M. Sullivan
- 8) M. Tihova, B. Tattre and P. Nicholls
- 9) V. Bolivar and R. Brown

LUNCH BREAK

2:00— 3:00 Session III. Chair: P. McGrath

- 10) C. Beaver and D.E. Mitchell
- 11) D.P. Phillips, J.M. Rappaport and J.M. Gulliver
- 12) W.K. Honig

COFFEE BREAK

3:30— 4:30 Session IV. Chair: V. Corkum

- 13) S. Nakajima
- 14) G. Eskes, M. Moscovitch and M. McCauley
- 15) P.D. Cole and W.K. Honig

WINE AND CHEESE

Thursday, September 10

9:00— 9:30 Coffee (Lounge)

9:30—11:00 Session V. Chair: S. Wright

- 16) J. Barresi and T. Juckes
- 17) C. Li and I.A. Meinertzhagen
- 18) V. Corkum and C. Moore
- 19) H. Piggins and B. Rusak

COFFEE BREAK

11:30— 1:00 Session VI. Posters (Lounge)

- 20) K. Davidson, W. Chaplin and K. Prkachin
- 21) P. McMullen
- 22) L. McAlpine and P. McGrath
- 23) J.F. Connolly and N.A. Phillips
- 24) G. Eskes and G. Craik
- 25) R. Braha and R. Klein
- 26) V. Bolivar, K. Manley, W. Danilchuk and J. Fentress

LUNCH BREAK

2:00— 3:00 Session VII. Chair: V. LoLordo

- 27) N.A. Phillips, C. Mate-Kole and R. Kirby
- 28) D.E. Mitchell and C. Beaver
- 29) D.P. Phillips and S.E. Hall

COFFEE BREAK

3:30— 4:30 Session VIII. Chair: C. Moore

- 30) M. Yoon and M. Yoon
- 31) E. Pyza and I.A. Meinertzhagen
- 32) K. Semba

PIZZA AND BEER IN THE LOUNGE

1. Patrick McGrath, G.Allen Finley, Gillian McNeill, Paula Forward and Peter Fitzgerald

PARENTAL MANAGEMENT OF POST-OPERATIVE PAIN

Although most children's post-operative pain is managed by parents at home, no studies have examined parental management of the adequacy of their practices. We will present data on 200 subjects from a survey of parents who have children aged 2-12 years who have undergone surgery and been discharged within a day. Data will be presented on levels of pain and treatment used for different groups.

2. Keith Jensen and John Fentress

THE DEVELOPMENT OF TRIADIC INTERACTIONS IN WOLVES (*CANIS LUPUS*) AND RED FOXES (*VULPES VULPES*)

Triadic interactions are social interactions that involve three individuals. They likely require greater social sophistication than dyadic interactions, and may be important for animals that live in complex social groups and engage in activities requiring coordination of behaviour. If triadic interactions are adaptive, then they should be exhibited more often and display a more coordinated form in social canids (wolves) than relatively asocial canids (foxes). Preliminary analyses indicate that during ontogeny, wolf pups do become more triadic in their social interactions than fox kits do. Further analysis will show whether the triadic interactions take on a more cooperative form in wolves.

3. Jeff Hamm

ORIENTATION EFFECTS DURING OBJECT NAMING

Are the effects of orientation on object naming due to finding the top of rotated objects or in identifying the object? Experiment 1 cued the location of the top prior to an object naming task. Experiment 2 cued object identity prior to a top-bottom discrimination task. Experiment 1 showed no influence of cuing, and Experiment 2 showed a benefit of cuing identity. Neither experiment showed a reduction in the effects of orientation due to cuing. The effects of orientation found in top-bottom discriminations may be explained as response competition.

4. Jenny Ryon, Paul Paquet and Richard Brown

WOLVES' RESPONSES TO CONSPECIFIC URINE-MARKS

We studied urine-marking and responses to marks in a pack of gray wolves (*Canis lupus*) housed at the Dalhousie Animal Behaviour Field Station for five years. Dominant wolves of both sexes urinated at higher frequencies than subdominant wolves. Responses included sniffing, over-marking, and licking. The type of response varied with the sex, age, reproductive status, and/or age of both the marking and the investigating individual. In adults, marks of female wolves were investigated more than those of males. The frequency of all types of response for the marks of adult females followed the same circannual rhythm of urine-marking frequency, with a peak during the winter mating season. Urine-marks may aid in reproductive synchrony in some wolves, while suppressing reproductive readiness in others.

5. Heather Schellinck¹, Cathy Smyth², Richard Brown^{1,2} & Michael Wilkinson^{2,3}
Departments of Psychology¹, Physiology and Biophysics² and Obstetrics and Gynecology³, Dalhousie University

EXPRESSION OF C-FOS IN THE ACCESSORY OLFATORY SYSTEM OF JUVENILE FEMALE MICE AFTER EXPOSURE TO MALE-SOILED BEDDING

Exposure to the urine or soiled bedding odours of novel adult males is known to accelerate puberty in juvenile female mice. To determine what part of the olfactory system is activated by these odours, the expression of *c-fos* in the main olfactory bulb (MOB) and the accessory olfactory bulb (AOB) of juvenile female mice was examined after their exposure to male-soiled bedding, peppermint extract or their own bedding. Well defined immunoreactivity was found throughout the AOB of the juvenile female mice exposed to the bedding odours of adult males for 3 and 12 hours. In contrast, dense fos staining was found in the granular cell layer of the MOB of mice exposed to peppermint odours for 3 or 12 hours. Of the mice exposed to their own bedding for 3 or 12 hours, none showed immunostaining for fos in the AOB; slight or no staining was observed in the MOB. These data indicate that social odours activate fos in the AOB while non-social odours activate the MOB.

6. April D'Aloisio, Mick Sullivan and Fionnula Murphy

DYSPHORIA AND CERTAINTY FOR SOCIAL JUDGEMENTS

Nineteen dysphoric and 20 nondysphoric subjects read brief person descriptions that were either high or low in information value, and were asked to make trait ratings for each person described. Subjects were then asked to indicate their level of certainty in the ratings they made. Results showed that dysphoric subjects were less certain than nondysphoric subjects in the trait ratings they made for low information descriptions. It is argued that, in the absence of detailed stimulus information, dysphoric individuals may be impaired in their ability to generate heuristic mental representations of social stimuli. The findings are discussed in relation to current positions on biased and realistic information processing in dysphoria.

7. Judith Francis and Mick Sullivan

DYSPHORIA AND APPLICATION OF GENDER STEREOTYPES

This present study investigated the relationship between mood and application of gender stereotypes. Dysphoric and non-dysphoric subjects read statements describing a male or a female target behaving in a manner consistent or inconsistent with stereotypes for the target's gender. A three-way interaction was found between mood (dysphoric, non-dysphoric), target sex and stereotype consistency of behaviour (consistent, inconsistent). Dysphoric subjects rated female targets as more likely to engage in stereotype consistent behaviour than non-dysphoric subjects. As a function of limited cognitive resources, dysphoric subjects may have used stereotypes as a heuristic to reduce the cognitive processing demands of the prediction task.

8. Mariana Tihova, B. Tattrie, and Peter Nicholls

FREEZE-FRACTURE ELECTRON MICROSCOPY OF RECONSTITUTED CYTOCHROME OXIDASE

Eukaryotic cytochrome oxidase (CO) is a multisubunit enzyme spanning the inner mitochondrial membrane. The size, shape and other gross features of the 3-dimensional structure of this protein have been obtained only for its isolated crystal form. The enzyme can be incorporated into liposomes of a defined lipid composition. Freeze-fracture electron microscopy of cytochrome oxidase vesicles (COV) was used to study the structure of the reconstituted enzyme. The enzyme incorporation was characterized by size distribution of the protein "bumps" of Pt/C replicas of convex and concave surfaces determined using a MCID program (Imaging Res. Inc.). Ta/W shadowing was used to obtain images with higher resolution, which allow us to find out some details on the surfaces of the protein.

9. Valerie Bolivar and Richard Brown

THE ONTOGENY OF ULTRASONIC VOCALIZATIONS IN CNS MYELIN DEFICIENT (JIMPY) MICE

We recorded both the ultrasonic vocalizations (UVs), at 65, 70, 75 and 80 kHz, and motor behaviours of jimpy mouse pups at 2 day intervals from days 2 to 20 postpartum. Jp/Y males vocalized less frequently than their normal male littermates from 2 to 8 days of age. For both jp/Y and normal mice, the rate of UVs was highest at day 2 and decreased with age up to day 8, after which UVs ceased almost completely. Our results indicate that UVs may be useful as an early predictor of future neuromuscular anomalies.

10. Chris Beaver and Donald E. Mitchell

PATTERN OF *C-FOS* EXPRESSION IN AREA 17 OF VISUALLY DEPRIVED KITTENS FOLLOWING INTRODUCTION OF NORMAL VISUAL INPUT

Last year we reported on the rapid expression of *c-fos* in various visual cortical areas of dark-reared kittens following their initial exposure to light. Here we focus on the pattern and time course of this expression visualized in both coronal sections and in unfolded and flattened sections of the visual cortex. Data relevant to two issues will be discussed, namely, the possible involvement of *c-fos* in the signal pathway that mediates experience-dependent change in the visual cortex during development, and possible modular organizational principles of the cat visual cortex.

11. Dennis P. Phillips, Jamie M. Rappaport and J. Mark Gulliver

ODE TO A HAIRDRESSER. WHY SHE SMILES AND NODS AT PARTIES, OR, WHEN "YES" MEANS "I DIDN'T UNDERSTAND A DAMNED WORD YOU JUST SAID".

One of the most common complaints of listeners with noise-induced cochlear hearing loss is that they have difficulty discriminating speech in noisy environments. Historically, this problem has been attributed to a loss of frequency resolving power in the cochlea. This is because the impaired frequency resolution is independently demonstrable, and because it is, in the principle, the kind of deficit that might prevent the ear from encoding the distributions of spectral energy that identify the phonemes of speech—a problem that is presumably only exacerbated by the presence of masking noise. In this talk, we present evidence which (a), knocked our socks off, and (b) suggests that a defect in temporal, rather than frequency, resolution accounts for the perceptual difficulty reported by these patients.

12. Werner K. Honig

DISCRIMINATION BY PIGEONS OF UNIFORMITY AND MIXTURE IN ARRAYS OF STIMULUS EFFECTS

Pigeons pecked at a video "touch screen" on which randomized arrays of 36 small squares of two different colors were presented in various proportions. With uniform arrays of red or blue elements, responding to the right side of the screen was rewarded. With all mixed arrays, containing both red and blue elements, responding to the left side was rewarded. The pigeons provided orderly proportions of choices in the form of gradients which reflect the degree of stimulus mixture in the arrays. This discrimination transferred readily to arrays of green and blue elements, and to arrays consisting of fewer (25) and more (64) elements than the training arrays.

13. Shinshu Nakajima

LOCATION OF DOPAMINERGIC SITES CRITICAL FOR BRAIN-STIMULATION REWARD

The locus of dopamine receptors critically involved in the generation of reinforcing effect of brain stimulation was examined.

Rats were trained to press a bar for the lateral hypothalamic stimulation, and D1 or D2 antagonists was injected through the cannula during the operant responding.

D1 antagonists suppressed responding when injected into a part of the nucleus accumbens, ventral tegmentum, or their vicinities. D2 antagonist suppressed responding only when injected into the nucleus accumbens, and not into any other loci.

14. Gail Eskes, Morris Moscovitch and Mary McCauley

**A PICTURE IS NOT ALWAYS WORTH A THOUSAND WORDS:
DISSOCIATIONS BETWEEN EXPLICIT AND IMPLICIT MEMORY
TESTS.**

Performance on explicit (e.g., free recall) and implicit (e.g., word stem completion) memory tests can be dissociated experimentally; implicit tests are said to involve automatic or unconscious memory processes, while explicit tests require strategic or consciously-controlled processes. Thus, it has been traditionally assumed that explicit tasks would benefit from elaborative processing during study (e.g., imagery), while this kind of processing would have no effect on implicit memory tasks. The current study suggests that while imagery does benefit explicit test performance as expected, using imagery during study can actually reduce implicit task performance in young adults. In contrast, older adults show reduced benefits of imagery on explicit tasks, and fewer costs of imagery on implicit tasks. Performance on some implicit memory tasks may not be so automatic, after all.

15. Patricia Cole and Werner K. Honig

**PIGEON UMWELT II: IS A PICTURE AS GOOD AS A PLACE TO A
PIGEON?**

Pigeons were trained on a simultaneous slide discrimination between colored slides of various views of the ends of a rectangular room. A control group was trained on an irrelevant slide discrimination between outdoor scenes. After reaching criterion on this "2D" task, the birds were released into the actual room previously represented in slide photographs and allowed to forage. Their behaviour in this "3D" task suggested the photographic medium may capture real world cues that are salient to pigeons as well as humans.

16. John Barresi and Tim Juckes

**MEDITATIONS ON MALCOLM X: A SOCIO-PSYCHOLOGICAL
ANALYSIS**

The relationship between individual and social forces in the historical transformation of society is not well understood. On the one hand, it appears that it is the action of individuals in pursuit of personal (subjective) goals that produces social change. On the other hand, it appears that (objective) social forces produce change. We (Juckes and Barresi, in preparation) have developed a model that articulates how these two types of forces can be understood jointly to produce social change. In the present paper the life-history of Malcolm X and his role in the development of black consciousness in America will be used to illustrate the usefulness of this model.

17. Chinglu Li and Ian A. Meinertzhagen

IN VITRO STUDY OF *DROSOPHILA* VISUAL SYSTEM

With its much simpler structural and functional organization compared to that of the vertebrates, the visual system of fruit fly, *Drosophila*, provides an ideal preparation for the studies of synaptogenesis. Success of *in vitro* culture of this system can further simplify this system, permitting experimental control over spatial and temporal aspects of its development. By screening different culture conditions, we have established primary cultures of eye imaginal discs and its target neuron, the optic neurons.

18. Valerie Corkum and Chris Moore

CUES FOR JOINT VISUAL ATTENTION IN INFANTS

Joint visual attention refers to the ability to follow another's direction of gaze. This project examined the relative importance of eye and head orientation in the establishment of joint attention in infants ranging from 6 to 19 months. In a face to face interaction with the experimenter, infants were exposed to four types of trials in which head and/or eye position varied: H (head only), E (eyes only), H+E (head & eyes match), and H-E (head & eyes mismatch). Results indicate that 6 to 10 month olds do not reliably engage in joint attention based on information from any of the cues while 12-13 month-olds establish joint attention based on head position alone. In contrast, 15 to 19 month olds rely on congruent head and eye position for determining direction of gaze and joining attention. At no age did infants establish joint attention based on eye position alone whether presented in isolation or in the context of the H-E cue. Based on these findings it appears that while head orientation is initially the most salient cue, by 15 months, head and eye orientation play equal roles in the establishment of joint attention.

19. Hugh Piggins and Ben Rusak

ELECTROPHYSIOLOGICAL INVESTIGATIONS ON THE SUPRACHIASMATIC NUCLEUS IN VITRO

The suprachiasmatic nuclei (SCN) of the hypothalamus house a dominant circadian pacemaker. Our research involves monitoring the effects of exogenously administered neurochemicals on the extracellularly recorded spontaneous firing activity of SCN cells maintained in a brain slice preparation *in vitro*. In particular, we examine how substances which are thought to be released from the visual projections to the SCN and substances which are intrinsic to the SCN affect SCN cells. Through this research, we aim to characterize the neuropharmacology of the circadian clock.

20. Karina Davidson, William Chaplin and Kenneth Prkachin

USER FRIENDLY STATISTICS FOR EVALUATING RISK FACTORS FOR CHD MORTALITY

Large scale studies have relied on tests of statistical significance to index and evaluate risk factors to coronary heart disease (CHD). Because such statistics are influenced by sample size and measurement error, they do not provide a clear indication of the practical implications of risk factors for mortality from coronary heart disease.

In this paper, we describe an alternative to traditional tests of statistical significance, Rosenthal's binomial effect size display (BESD), that effectively and unambiguously communicates the pragmatic importance of risk factors for coronary heart disease (CHD) mortality. Moreover, the BESD facilitates both the evaluation of the relative importance of risk factors within one study, and the stability of the same risk factor in different studies.

21. Patricia A. McMullen

NAMING FAMILIAR FACES ROTATED IN TWO AND THREE DIMENSIONS

Naming latencies have supported mental representations of objects in 3/4 view. Yet, cells in the monkey temporal gyrus respond preferentially to frontal and profile views of faces. Are objects and faces represented in fundamentally different ways? Frontal, 3/4 and profile views of familiar faces were named. Like objects, upright faces in 3/4 view were named the fastest. However, rotations in the picture-plane had equivalent effects on latencies for 3/4 and frontal views and much greater effects for profile views.

22. Linda M. McAlpine and Patrick J. McGrath

A YOUNG CHILD'S DEVELOPING UNDERSTANDING OF PAIN

Further advancement in assessment and treatment of pain in children has been hampered by our limited knowledge of emerging concepts of pain in preschool children and by their immature expressive language. Methodologies used to research a child's developing "theory of mind" proved useful in revealing how young children conceptualize pain. It was hypothesized that preschool children would progress from recognizing pain in others to imputing to others the emotions which pain evokes and subsequently to predicting behaviour resulting from pain experience. This hypothesis was investigated through collection of a two week language sampling from subjects (30 children, aged 18-76 months) augmented by data collected during structured play sessions in which the children assume the characters in pictures and of toy figures who experience pain. Comparisons were made to the children's developing concepts of internal states in themselves and others.

23. John F. Connolly and Natalie A. Phillips

EVENT-RELATED POTENTIALS TO PHONOLOGICAL AND SEMANTIC PROCESSING IN SPOKEN SENTENCES

In previous studies (Connolly et al., 1990; 1992) a negative-going component peaking at about 200-250 msec after the onset of the terminal word of a low contextually constrained (spoken) sentence was observed in conditions which also elicited a N400 response. The N200 and N400 were found to be differentially sensitive to phonologically correct masking—N400 peak latency was delayed while N200 was unaffected. The N200 was hypothesized to be sensitive to the initial phoneme of the terminal word and a reflection of the acoustic-phonetic process in the phonological stage of word processing. The present experiment was designed to explore further the N200 and the N400 elicited by the terminal words of spoken sentences. Four conditions were employed which elicited the N200 and N400 together, separately and not at all. Results are interpreted as supporting the idea of the N200 as reflecting an acoustic-phonetic process and the N400 as a process dependent upon the "outcome" of the process(es) manifested by the N200. Results for the N200 are also interpreted in relation to the cohort model of spoken word recognition with particular reference to the concept of recognition-point.

24. Gail Eskes and Gus Craik

MEMORY AND AGEING: HOW DOES SEMANTIC PROCESSING HELP?

While there is general agreement that semantic processing during item study enhances later recall, little attention has been directed at investigating whether different types of semantic processing (e.g., cueing superordinate vs subordinate properties) are differentially effective in increasing memory performance. The results of this study suggest that some types of semantic processing can have differential effects on memory performance. In general, however, processing of superordinate or subordinate features had similar effects on memory, while words associated with more elaborative processing were better recalled by both young and older adults.

25. Richard Braha and Raymond Klein

VISUAL WORD RECOGNITION AND VISUAL ATTENTIONAL PROCESSES IN DEVELOPMENTAL DYSLEXIA: A FAMILIAL CASE STUDY*

Reading and visual attentional processes were studied in a family with two known developmentally dyslexic children. The family underwent neuropsychological assessment and a series of cognitive experiments. Control group S's provided normative data. The sets included: Coltheart's reading battery; a test of memory for visual spatial and temporal information; Olson's component reading skills task; visual attentional distractor tasks, and 4 separate tests of exogenous and endogenous control over overt and covert orienting of visual attention. Both parents and 4 of the 5 children displayed reading disabilities and several similar problems with visual attentional processes.

* Previously presented at the Rotman Research Institute's conference on attention (Toronto, March, 1992).

26. Valerie Bolivar, Kevin Manley, Wanda Danilchuk and John Fentress

DEVELOPMENT OF MOVEMENT SEQUENCES IN JIMPY (CNS MYELIN DEFICIENT) MICE

We examined the ontogeny of grooming and swimming behaviors in mutant jimpy male pups and their littermate controls. The animals were videotaped at 2 day intervals from the 3rd to the 21st day postpartum. Developmental measures included the percent time spent grooming, the number of grooming bouts and interlimb coordination during swimming. Results indicate that the mutants displayed a developmental lag in swimming ability, however, they did spend more time grooming and groomed earlier than littermate controls. Our data confirms the importance of detailed behavioural documentation if we are to understand how gene expression affects motor development.

27. Natalie A. Phillips, C. Charles Mate-Kole, and R. Lee Kirby

COGNITIVE IMPAIRMENT IN PATIENTS WITH LOWER EXTREMITY AMPUTATIONS SECONDARY TO PERIPHERAL VASCULAR DISEASE

Peripheral vascular disease (PVD) refers to disease of any of the blood vessels outside the heart and accounts for approximately 75-80% of lower-extremity amputations performed today. Although patients with PVD are considered neurologically normal, PVD might lead to cerebral ischemia or cerebrovascular disease. We hypothesized that patients with PVD would show cognitive deficits on neuropsychological assessment. Memory, language function, visuospacial ability, and problem solving were evaluated in 14 patients with lower-extremity amputations secondary to PVD and 16 healthy elderly age-matched controls. Patients with PVD performed more poorly than controls on tests of visuospacial ability, problem solving and abstract reasoning. It is suggested that the cognitive status of amputee patients with PVD should be considered in prosthetic rehabilitation programs.

28. Donald E. Mitchell and Chris Beaver

DEVELOPMENT IN KITTENS BASED ON OBSERVATIONS OF CAT-301 IMMUNOREACTIVITY IN THE dLGN OF VISUALLY DEPRIVED ANIMALS

The monoclonal antibody, Cat-301, recognizes a surface associated antigen (a chondroitin sulphate proteoglycan) on a specific class of neurons (Y-cells) in the central visual pathways of cats. Cat-301 is first expressed in the dLFN of kittens at 4 weeks of age and adult levels are not reached until 6 months of age, a time course that is mirror symmetric to the declining ability of the central visual pathways for experience-dependent change. During the course of experiments that explored further the experience-dependent nature of Cat-301 expression, we observed that short periods of monocular deprivation resulted in low overall immunoreactivity in both the deprived and nondeprived laminae of the dLDN's. This and other findings from this investigation hold implications for the timing sensitive periods in visual development.

29. Dennis P. Phillips and Susan E. Hall

SELECTIVE ADAPTATION STUDIES OF CAT CORTICAL AUDITORY NEURONS

Studied with tone pulses of any given amplitude, neurons of the cat's primary auditory cortex are excited by a narrow range of tone frequencies. Curves plotting response rate as a function of frequency have an inverted-V shape, and are very narrow in relation to the animal's audible range. On the basis of persuasive, if somewhat thin evidence, we once argued that in some neurons, such a curve actually represents the envelope of the response ranges of a number of independent inputs tuned to similar, but not necessarily identical, frequencies. We have now directly tested this notion by obtaining response-vs-frequency functions for tone impulses in the absence and presence of a continuous tonal masker whose frequency was set at the edge of a neuron's effective frequency range. Our data support the hypothesis that the response-vs-frequency function is indeed, for some neurons, the envelope of those for a number of independent, closely-tuned points.

30. Myong G. Yoon and Myonghae Yoon

ACTION DEVELOPS EARLIER THAN ATTRIBUTION OF THINKING OR RETROACTIVE JUSTIFICATION

We studied the development of children's understanding of false beliefs in cross-sectional (3y1m - 5y0m; N=88) and longitudinal (2y8m - 4y6m; N=11) experiments. During 3y1m - 3y6m, 52% of 22 children were able to make correct predictions such that the protagonist's action would be consistent with the protagonist's false belief. Most (95%) of them, however, were unable to explain their own correct predictions, and rarely made an explicit attribution of the false belief to the protagonist. They also showed difficulties in justifying the protagonist's futile action retroactively, in spite of their preceding successful predictions of the protagonist's action. During 3y7m - 4y0m, 38% of 22 children acquired new abilities to explicitly attribute the false belief to the protagonist, whereas they made only a minor improvement in making correct predictions (61%). In general, children's ability for a correct prediction of action developed 4-7 months earlier than those for explanation of prediction, attribution of thinking, or retroactive justification of the protagonist's futile action.

31. Elzbieta Pyza and Ian A. Meinertzhagen

DAY/NIGHT AND CIRCADIAN RHYTHMS IN THE FIRST VISUAL NEUROPILE IN THE OPTIC LOBE OF THE FLY *MUSCA DOMESTICA*

In the fly's visual neuropile, or lamina of the optic lobe, the frequency of synaptic profiles changes during the 24 hours period. Under light/dark conditions (LD 12:12), the number of afferent tetrad synapses formed upon two interneurons L1 and L2 is higher during the day but number their reciprocal counterparts from L2 formed back upon receptor terminals is higher during the night. Examination of the presynaptic ribbon profiles under constant darkness conditions (DD) revealed a strong circadian rhythm in frequency of feedback synapses and less pronounced rhythm in frequency of tetrad synapses.

In the present study, the size of presynaptic ribbon profiles in single EM sections has been measured at LD 12:12 conditions. Moreover, we have measured area, perimeter, length, formfactor and diameter of two interneurons L1 and L2 under LD and DD conditions. The results obtained from this study have shown no statistically significant differences in the size of presynaptic ribbon of tetrad and feedback as well as synapses. In parameters of L1 and L2 interneurons, however, there were remarkable differences at all levels of lamina depth between day and night at LD and between subjective day and subjective night at DD conditions. L1 and specially L2 interneurons were bigger during the day and subjective day than during the night and subjective night.

32. Kazuo Semba

CHOLINERGIC AND SEROTONERGIC MECHANISMS OF RAPID EYE MOVEMENT (REM) SLEEP

Cholinergic neurons in the mesopontine tegmentum have implicated in the regulation of REM sleep. Serotonergic suppression of these cholinergic neurons has long been postulated on the basis of behavioural studies, but no direct proof was available. We have investigated serotonergic input to cholinergic neurons using both physiological (intracellular recordings using brain slices in vitro) and anatomical (tract tracing, immunohistochemistry, and electron microscopy) techniques. The results were consistent with the suggested role of serotonin in modulating rapid eye movement sleep phenomenology through its inhibitory action on cholinergic neurons in the mesopontine tegmentum.